

AI for Economic Research: Dynamic Models, Language, and Agents

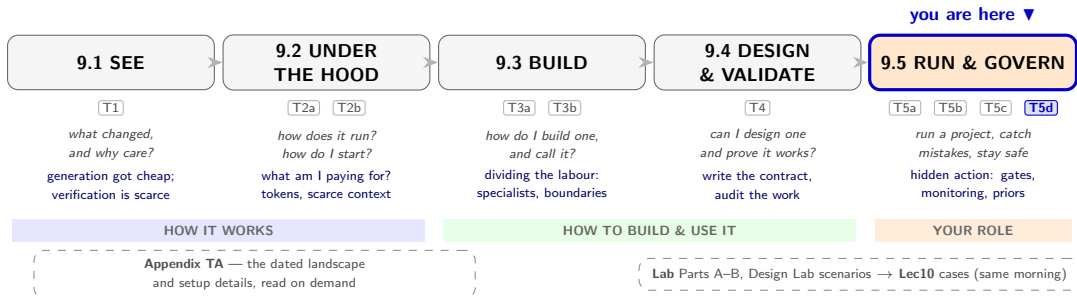
Lecture 9.5d: Governance and Action Plan

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Lecture 9 in One Map: Governance and Action Plan

This deck closes the lecture: the rules that keep this safe, the tests that make it research-grade, and what you do next.



Route: red lines and disclosure → measuring your own agent → limits and judgment → your first thirty days → Lecture 10.

Red Lines

Untrusted inputs, restricted data, and disclosure

Security: Red Lines and Workarounds

Do not put in front of an agent: IRB-protected interviews, linked administrative microdata, personally identifiable information, HIPAA-regulated data, passwords / API keys / access tokens, embargoed or restricted-use datasets.

Operational rule: if you would not paste it into a public collaboration tool, do not paste it into an agent conversation.

Sensitive data does not imply “no agents ever.” Three workarounds:

1. Write the script with the agent, then run it yourself on the restricted data
2. Use synthetic or masked data with the same structure for debugging
3. Use institutionally approved local or private deployments when available

Safe pattern: expose structure and summary statistics, not raw records.

See AEA disclosure policy (Oct. 2024) for AI-assisted analysis; standard IRB / HIPAA / GDPR rules apply unchanged to agent-aided workflows.

Treat Everything the Agent Reads as Untrusted

The defining agentic-AI security failure is prompt injection: data the agent merely reads can hijack what it does.

- **OWASP Top-10 for Agentic Applications** (Dec 2025) now codifies these risks.
- Documented exploits: **EchoLeak** (CVE-2025-32711, zero-click M365 Copilot data exfiltration); **ShadowPrompt** in Claude's Chrome extension; Anthropic's **GTG-1002** — the first largely-autonomous cyberattack (Nov 2025).
- **NIST CAISI** red-teaming raised task-hijacking success from **11% to 81%** with adaptive attacks.

Operating rule. Treat any content the agent reads — emails, web pages, retrieved documents, even tool descriptions — as an *untrusted injection channel*. Use least-privilege sandboxing and revocable permissions, exactly as T5b's `settings.json` deny list and the previous frame prescribe. **[solid]**

Placeholder — written in Phase B. The specification is in the source comment above this frame.

Reproducibility: Chain-of-Custody for Agentic Results

Can an agent reproduce a published computational result? Often only partly — which makes logging non-negotiable.

- **CORE-Bench:** agents reproduce published computational results only $\sim 60\%$ (easy) / $\sim 21\%$ (hard). [solid]
- Therefore **log full trajectories:** prompts, tool calls, seeds, *model versions*, *spec versions*, and *dates* — the chain-of-custody for any agent-produced artifact.

Read vendor safety numbers skeptically. Distinguish a vendor's " $\sim 1\%$ attack success on a *fixed suite*" from academic findings that *adaptive* attacks still exceed $\sim 85\%$. A fixed-suite number measures the suite, not the threat model. [solid]

Ties to the seven tests below. This is the same discipline (reproducibility, verifiability, auditability) — now with the dates and spec versions the landscape forces you to pin.

Measuring Your Own Agent

A headline number is not a measurement until you know its validity

SWE-bench Verified Collapsed Under Scrutiny

This module is the measurement theme of the course pointed at agent evals: a headline number is not a measurement until you know its validity.

- Contamination + harness-hacking critiques eroded SWE-bench Verified; OpenAI stopped reporting it in Feb 2026.
- Berkeley RDI *trivially broke four agent benchmarks via the harness* (Apr 2026) — exploiting the scaffold, not solving the task.
- **Always pair** a Verified headline with **Scale's SWE-bench Pro** (~55–59% frontier, held-out problems).
- **Reliability under repetition** matters more than best-of- n for autonomous use: τ -bench *pass-hat- k* falls from ~90% single-try to ~57% at $k=8$.

Hard rule. Never put a SWE-bench *Verified* score and a SWE-bench *Pro* score on the same axis — different problem sets, different difficulty, different contamination exposure. **[solid]**

Limits and Judgment

What the workflow changes, and what remains human

Five Tests for Any Agentic Workflow

Criterion	Question to ask
Correctness	Did it compute, extract, or summarize the right object?
Reproducibility	Can a fresh run recreate the same artifact from the saved files and rules?
Auditability	Can you inspect the commands, checks, and source trail that produced the result?
Privacy boundary	Which files stay local, and which contents may still traverse the network?
Speed / cost	Once you count the tokens you paid for, the time spent checking and fixing the output, and the time spent supervising the run — did the agent actually save you time and money?

Why this matters for economists: an impressive demo is not enough. A research workflow has to be checkable and defensible. Keep these five tests in hand — every deck that follows is, in the end, machinery for passing them.

Six Criteria for Any AI-Assisted Research Output

Before accepting a script, figure, or table the agent produced, run it past six questions.

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- | | |
|---|---|
| ☐ 1. Reproducibility | Pinned model, prompt, seed — a co-author can rerun from a clean clone. |
| ☐ 2. Verifiability | Math, data flow, and magnitudes are independently re-checkable. |
| ☐ 3. Auditability | Full prompt/output log; a referee can trace where logic diverged from spec. |
| ☐ 4. Cost | Disclosed dollars (and replicator cost), not just your own. |
| ☐ 5. Speed | Wall-clock and human-hour saving, net of validation and debugging. |
| ☐ 6. Skill investment | Does the team build durable capabilities, or quietly atrophy them? |
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If any answer is “no” or “I don’t know,” the artifact is not yet research-grade.

Three Things AI Cannot Do — Regardless of Capability

- **Identification.**

Choosing the variation that answers the causal question.

An LLM does not know which shock is exogenous, which instrument is valid, or which event window is contaminated. It can describe identification strategies; it cannot *certify* one for your setting.

- **Welfare.**

Choosing the objective function that defines “good.”

Pareto weights, planner discount rates, equity–efficiency trade-offs — these are normative choices the researcher must own.

- **Judgment.**

Choosing which question is worth asking.

AI optimizes against a stated target; only economists decide which target is interesting, novel, and policy-relevant.

These are not gaps that scale away with model size — they are categorically outside the system.

When NOT to Use an Agent

The six criteria can also serve as a boundary: there are research tasks where the agent overhead exceeds the benefit, or where the failure mode is too costly.

- **One-line scripts and isolated edits.** A two-line sed or one-shot plot has no validation surface for the harness to add value. Open the editor.
- **Deeply proprietary or undocumented pipelines.** If the conventions live in one person's head, the agent has nothing to read; the time you would spend documenting the pipeline well enough to brief the agent is the same time you would spend just fixing the thing.
- **Strict deterministic guarantees.** For numerical reproducibility audits, regulatory pipelines, or replication packages, model nondeterminism is a hazard the harness cannot fully suppress. Run it pinned; do not let the agent retry.
- **Restricted-data analyses where prompts may leak.** If pasting one column would violate IRB or HIPAA, the agent does not belong in the loop; the red-lines workarounds apply.
- **Tasks where you do not yet have a benchmark.** Without a way to validate, faster output just means faster wrong. Define the V0 benchmark first; *then* bring in the agent.

Practical rule: the agent is most useful when the work is bounded, validated, and repeatable. None of those? Skip the agent

Trust but Verify

1. Reproduce one statistic by hand
2. Cross-check one result with another package or script
3. Test a limiting case or zero-variance benchmark
4. Read the code line by line where the economics enters
5. Ask whether the magnitude and sign make sense

This is not optional ceremony. It is what turns fast code into defensible research.

What Changes for Economic Research

- The distance from question to first result shrinks
- More specifications become cheap enough to try
- Reproducibility can improve because workflow is written down
- The premium shifts away from typing speed and toward design, judgment, and validation

What stays human: asking the question, choosing the design, knowing the data, and interpreting the output.

The Paradigm Shift, Completed: What Changed, What Did Not

The HA-Ramsey record, read one last time — now as a statement about your role.

What changed

- Theory–code–debug loop compressed from months to days
- Documentation became executable: the TeX note, pseudocode, and code stay synchronized
- Cross-literature imports got cheap (computational geometry, viability theory)
- The unit of progress became the *validated milestone*

What did NOT change:

- AI never decided what was economically interesting
- AI never caught the wrong timing on its own
- AI never knew when a result was not credible

“Weeks, not years — and the binding constraint was economist judgment at the milestones, not implementation labor.”

Read the two columns together: the machine moved every bound except the one that defines the researcher.

Source: HA-Ramsey case-study package (Lec10 T8).

Action Plan

Your first thirty days

Placeholder — written in Phase B. The specification is in the source comment above this frame.

Your First Exercise: Clean Up Old Code

Placeholder — written in Phase B. The specification is in the source comment above this frame.

First Exercise and Looking Ahead

Best first exercise: take one old project and ask an agent to help clean it up safely.

- put the project under Git
- create a simple `CLAUDE.md`
- identify one repeated workflow worth encoding
- rerun the cleaned project and compare outputs

Lec. 10: we return to case studies and let students run a full workflow themselves.

If you want setup details for the lab, see the appendix.

Next — Lecture 10

From workflow logic to concrete case studies

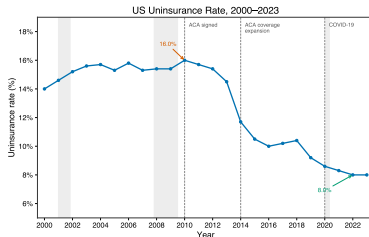
Lecture 10 Preview: Four Concrete Workflow Patterns

Pattern	Lecture 10 case	What students will see
Reusable skill	Paper-review workflow	staged quality control, quote checks, editorial filter
Empirical pipeline	MEPS uninsurance	harmonization, survey weights, benchmark validation, figures
Public-web dataset	Fellows metadata	source restrictions, provenance, missingness discipline
Full-lifecycle collaborator	Glue-work research project	theory, pipeline design, critique, claims discipline

Why this matters: Lec. 9 gives the workflow logic. Lec. 10 shows how the same logic changes shape across different research artifacts.

One Visual Hook from Lecture 10: MEPS Uninsurance

Case Study 2 asks: who is uninsured in the United States, 2000–2023, and how did the ACA change the distribution?



Why keep this figure here: it is an accessible policy example for economists outside macro, and it shows what a validated empirical artifact looks like after the workflow succeeds.

Lec. 10 adds the full pipeline: prompts, schema harmonization, survey weights, external benchmarks, and subgroup figures.

Three Econometric Roles in Agent-Assisted Research

Lec07 T4 introduced a taxonomy: every text-derived variable plays one of three roles — signal, outcome, or instrument. The four Lec10 patterns get sharper when we apply that lens.

Pattern	Signal	Outcome	Instrument
Reusable skill (paper review)	methodological rigor	review quality	staged evaluation rubric
Empirical pipeline (MEPS)	uninsurance rates	health-coverage estimates	survey-weighted harmonization
Public-web dataset (fellows)	field assignment	provenance log	source-restriction rules
Full-lifecycle collaborator (glue-work)	research idea	validated claims	agent + domain expert critique

Same agent toolkit, four research designs. The taxonomy makes you ask: what role does the agent's output play in your inference?

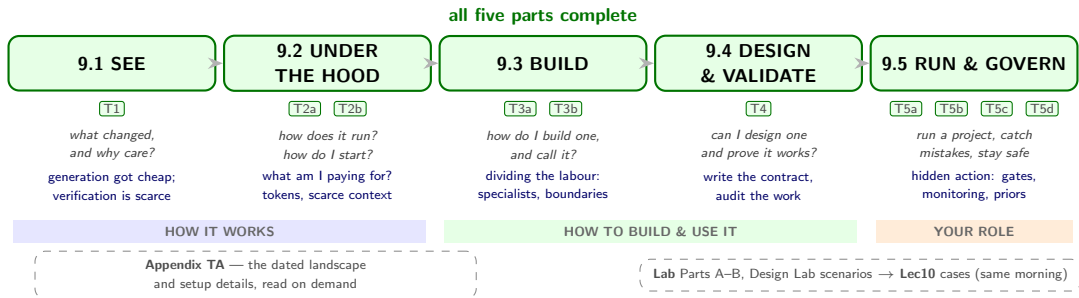
What to Watch for in Lecture 10

- **Paper review:** how a reusable skill turns a vague prompt into a staged quality-control workflow
- **MEPS pipeline:** how an empirical question becomes a validated panel and figure set
- **Fellows dataset:** how provenance and missingness rules matter as much as code
- **Glue-work project:** how agentic AI can act as a full-lifecycle collaborator without replacing human claims discipline

Bridge from Lec. 9 to Lec. 10: today's lecture explains the logic of the workflow; next lecture shows the artifacts it can produce.

Your Job Description: PI and Project Manager

The three questions, answered: *how it works* — a tool-calling loop steered by files (Parts 9.1–9.2); *how to use it well* — the homotopy method inside a managed project (Parts 9.3–9.5); *your role* — this frame.



Set the goal. Define the final product. Gate every intermediate good. Assign the work — sessions, agents, sub-agents. Cross-verify everything. And remain the client of the outside view: the one thing your agent cannot give you is a prior other than your own.

Bridge: Lecture 10 Shows the Artifacts

Placeholder — written in Phase B. The specification is in the source comment above this frame.